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YOUVA

• Medium • #3 • Longest Substring Without Repeating Characters

▲ Problem Statement • Given a string, find the length of the longest substring without repeating characters.

▲ Main Solution • Use an array to store the last index where each character appeared. If a character appears again inside the current substring, move "left" directly after its previous position.

▲ Brute Force • Check every possible substring & find the longest one without duplicate characters.

▲ Algorithm • 1] • Create an array "last [256]" & fill it with "-1".

2] • Set "left = 0" & "maxLength = 0".

3] • Loop through the string using "right".

4] • If the current character was seen inside the current window, move :-

"left = last [c] + 1;"

5] • Update the character's last position.

6] • Update "maxLength".

7] • Return "maxLength".

▲ Important Points • • "last [256]" stores the last position of each character.

• "-1" means the character has not appeared yet.

- If a duplicate is inside the current window ; jump " left " forward.
- Window Length :- " right - left + 1 "

▲ Complexity → Brute Force → $O(n^2)$ & $O(n)$
Optimized → $O(n)$ & $O(1)$

- **Medium** → #5 → Longest Palindromic Substring.

▲ Problem Statement → Given